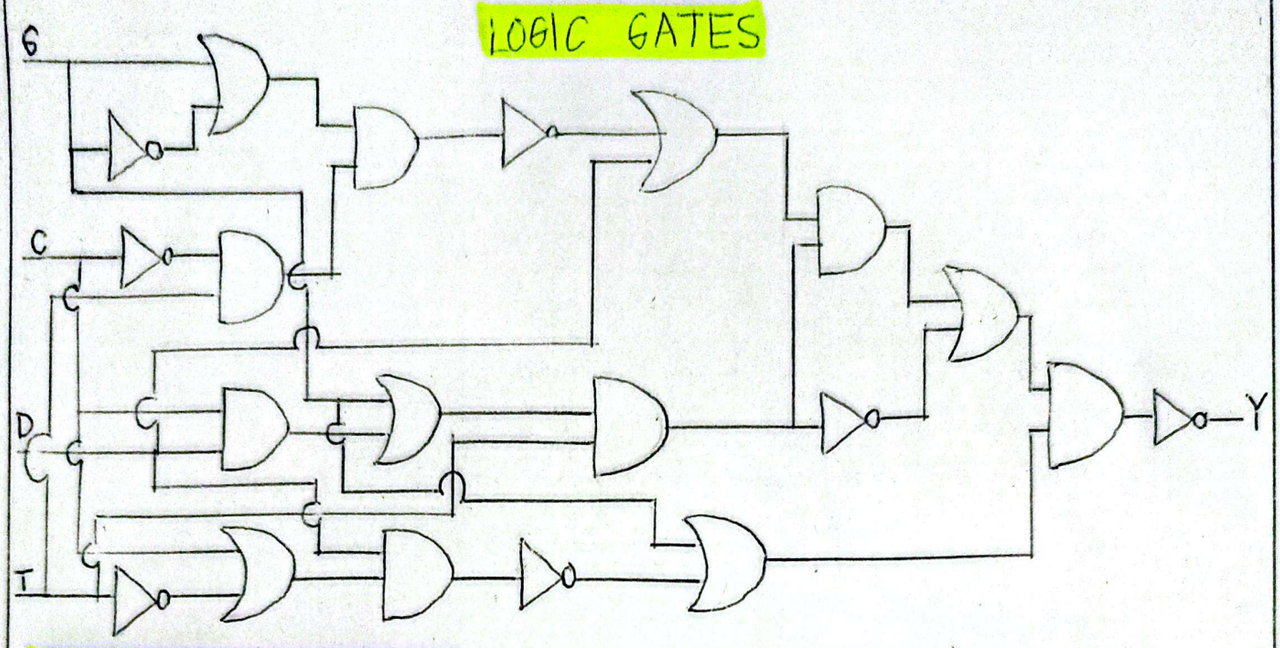
Gian Carlo D. Trilles

## LOGIC GATES




## BOOLEAN EXPRESSION

$$Y = \overline{[((\overline{T}+C)\cdot\overline{D})+G][(((C\cdot D)+G)\cdot T)\cdot\overline{(((G+\overline{G})\cdot(\overline{C}T))+D)+\overline{(((C\cdot D)+G)\cdot T)}]}}$$

## TRUTH TABLE

| G | C | D | T | Y |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 |

| SUBMITTED BY | SUBMITTED TO | |
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